

Local Model Bench

Which Studio model can be trusted with TyLog coding work

Runs 2026-10-09 to 2026-10-10 · base commit 9dbfa0f · data: tool/model_bench/results.jsonl

Ornith 1.5 35b passed 14 of 15 runs and finished every small task in about half a minute. It is the only model that fixed the real scanner bug more than once. No model was free of wrong work that looked finished, so every delegated diff still needs a human or codex review. The Ornith 27b coder result is a tooling failure, not a measure of that model.

ORNITH 35B PASSED
14/15
best of four

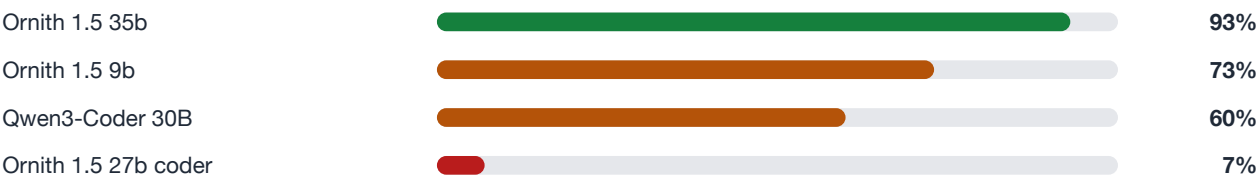
RUNS SCORED
60
4 models × 5 tasks × 3

WRONG BUT LOOKED
DONE
6
across three models

CONTROL RUNS
CORRECT
11/11
re-run 2026-10-10

Result

Model	Passed	Wrong but looked done	No change made	Touched other files
Ornith 1.5 35b	14/15	1	0	0
Ornith 1.5 9b	11/15	2	1	1
Qwen3-Coder 30B	9/15	3	1	2
Ornith 1.5 27b coder	1/15	0	13	0



“Wrong but looked done” counts runs where the model changed code, its own verify command passed, and the hidden check failed. It is the figure that decides trust: such work would be merged if nobody read the diff.

Passes by task

Task	Ornith 35b	Ornith 9b	Qwen3-Coder	Ornith 27b coder
New file from a spec	3/3	3/3	1/3	1/3
Same-value edit, 144-line widget	3/3	3/3	2/3	0/3
One branch in a 2,500-line file	3/3	1/3	3/3	0/3
Real bug fix in scanner.dart	2/3	1/3	0/3	0/3
Write a guard test	3/3	3/3	3/3	0/3
Total	14/15	11/15	9/15	1/15

Median seconds per run

Task	Ornith 35b	Ornith 9b	Qwen3-Coder
New file from a spec	33	57	25
Same-value edit	28	24	190
One branch in a large file	22	16	121
Real bug fix	806	901	1,927
Write a guard test	34	78	160

Medians are used because two Ornith 9b runs (5,697 s and 553 s) were stretched by the laptop sleeping. Their pass or fail outcome is unaffected. The Ornith 27b coder is left out: its runs ended in 3 to 40 seconds without doing the work.

Findings

Ornith 1.5 35b	Clean on all four small tasks; fixed the bug in 2 of 3 runs	Use as default
Ornith 1.5 9b	Follows a spec; 1 of 3 inside a large file	Small files only
Qwen3-Coder 30B	3 of 3 in the large file; drops rules from a spec	Pinpoint edits only
Ornith 1.5 27b coder	Stops before editing; tool-calling fault not investigated	Not measured

- **Ornith 35b** made no error on the 12 small-task runs. Its single failure was on the bug fix: the targeted tests passed and two donor-adoption tests elsewhere in the core suite broke.
- **Ornith 9b** failed twice on the large-file edit, once printing the literal text `<code>` instead of the status code and once making no change, and once touched a file outside its allowlist.
- **Qwen3-Coder** passed the new-file task once in three; both failures ignored the rule to keep the detail suffix and still passed lint. On the same-value edit it changed a 2 into a 4. It never fixed the bug and twice edited files it was told to leave alone.
- **Ornith 27b coder** made no change in 13 of 15 runs. In the one log read in full it located the function, announced the edit and ended the turn without calling the edit tool. The cause was not investigated.

Recommendation

Make Ornith 1.5 35b the default Studio model for delegated work, including small edits to existing files that were going to codex. Keep codex for multi-file and design-heavy fixes. Keep reading every diff: the best model still produced one wrong result that passed its own check.

Method

Five tasks were cut from this repository’s own history, so each has a known-good answer. Every run starts from commit `9dbfa0f` in a separate git worktree. The model works through `pi-run.sh` with a visible `verify` command and up to three attempts of 900 seconds. The score comes from a check the model never sees.

Task	Visible to the model	Hidden check
New file	lint of the new file	unit test over the full spec table
Same-value edit	lint of the file	every number keeps its value; at least 3 tokens used
Large-file edit	lint of the file	unit test; at most 3 lines deleted; file not truncated
Bug fix	the failing cache tests	whole core suite passes; tests not edited

Task	Visible to the model	Hidden check
Write a test	the new test passes	test fails when either entitlements file is broken

Controls

`selftest.sh` checks the bench itself before any model is scored. All 11 control runs gave the required outcome when re-run for this report.

Control	Input	Required	Result
Null	untouched baseline, 5 tasks	fail	5/5 failed
Reference	the real fix from git history, 5 tasks	pass	5/5 passed
Sloppy	same-value edit with one number changed	fail	1/1 failed

Limits

- Three runs per cell. A 2/3 against a 3/3 is one run of difference; only the larger gaps are meaningful.
- One harness (pi) and one prompt per task. A model that calls tools differently, as the 27b coder appears to, is scored as failing.
- Lint is a weak visible check on three tasks by design, to expose work that looks finished. With a stronger visible check some failures would be retried.
- All models ran on the Studio PC, one at a time. Times include model loading and the local test runs on the laptop.
- Reproduce with `tool/model_bench/run_all.sh`; the table with `report.py`.